

Curriculum Vitae

Personal Data

Title	Dr.-Ing.
First name	Ali Ahmad
Name	Khalifa
Current position	Postdoctoral Researcher
Current institution(s)/site(s), country	Chair of Process Systems Engineering, Technical University of Munich, Freising, Germany
Identifiers/ORCID	0000-0002-4822-3436; Google Scholar

Qualifications and Career

Stages	Periods and Details
Degree programme	B.Sc., Mechanical Engineering, 09.2008–01.2013. University of Jordan, Amman, Jordan. M.Sc., Chemical and Energy Engineering, 10.2013–04.2016. Otto von Guericke University, Magdeburg, Germany.
Doctorate	Oral examination: 19.09.2022; with distinction. Supervisor: Prof. Michael Breuer. Subject: Modeling and simulation of the breakage of cohesive particle agglomerates in turbulent wall-bounded flows. Helmut-Schmidt University, Hamburg, Germany.
Academic / professional career	Postdoctoral Researcher, 11.2025–present: Technical University of Munich, Chair of Process Systems Engineering. Postdoctoral Researcher, 10.2022–10.2025: Helmut-Schmidt University, Department of Fluid Mechanics. Doctoral Researcher, 12.2016–09.2022: Helmut-Schmidt University, Department of Fluid Mechanics. Student Research Assistant, 12.2015–10.2016: Otto von Guericke University, Laboratory of Fluid Dynamics and Technical Flows.

Activities in the Research System

Lecturer at TUM: Control Engineering (summer semester 2026); Machine Learning in Food and Life Sciences (winter semester 2026/27, scheduled). Previously Teaching Assistant in Fluid Mechanics and Applied Fluid Mechanics at Helmut-Schmidt University.

Supervision of student research projects and master's theses at Helmut-Schmidt University and TUM; supervision of a DFG-funded student research assistant at TUM; contribution to doctoral supervision in Hamburg. Co-PI of DFG project 573724408 (KH 653/3-1; 12.2025–11.2028).

Scientific Results

Category A

1. **A. Khalifa**, M. Breuer and H.H. Nguyen, 2026. Assessment of Euler–Lagrange models for fluid-induced cohesive particle deagglomeration using particle-resolved DNS. *Particuology* 114, 408–430. DOI: 10.1016/j.partic.2026.04.022
2. **A. Khalifa** and M. Breuer, 2025. Investigation of cohesive particle deagglomeration in homogeneous isotropic turbulence using particle-resolved direct numerical simulation. *Physics of Fluids* 37, 083339. DOI: 10.1063/5.0278786
3. **A. Khalifa** and M. Breuer, 2024. Evaluation of an efficient data-driven ANN model to predict agglomerate collisions within Euler-Lagrange simulations. *Computers & Fluids* 269, 106119. DOI: 10.1016/j.compfluid.2023.106119
4. **A. Khalifa** and M. Breuer, 2023. Data-driven ANN approach for binary agglomerate collisions including breakage and agglomeration. *Chemical Engineering Research and Design* 195, 14–27. DOI: 10.1016/j.cherd.2023.05.040
5. **A. Khalifa**, M. Breuer and J. Gollwitzer, 2022. Neural-network based approach for modeling wall–impact breakage of agglomerates in particle–laden flows applied in Euler–Lagrange LES. *Int. J. Heat Fluid Flow* 94, 108897. DOI: 10.1016/j.ijheatfluidflow.2021.108897
6. **A. Khalifa**, J. Gollwitzer and M. Breuer, 2021. LES of particle-laden flow in sharp pipe bends with data-driven predictions of agglomerate breakage by wall impacts. *Fluids* 6 (12), 424. DOI: 10.3390/fluids6120424
7. **A. Khalifa** and M. Breuer, 2021. An efficient model for the breakage of agglomerates by wall impact applied to Euler–Lagrange LES predictions. *Int. J. Multiphase Flow* 142, 103625. DOI: 10.1016/j.ijmultiphaseflow.2021.103625
8. **A. Khalifa** and M. Breuer, 2020. Data-driven model for the breakage of dry monodisperse agglomerates by wall impact applicable for multiphase flow simulations. *Powder Technology* 376, 241–253. DOI: 10.1016/j.powtec.2020.07.089
9. M. Breuer and **A. Khalifa**, 2019b. Refinement of breakup models for compact powder agglomerates exposed to turbulent flows considering relevant time scales. *Computers & Fluids* 194, 104315. DOI: 10.1016/j.compfluid.2019.104315
10. M. Breuer and **A. Khalifa**, 2019a. Revisiting and improving models for the breakup of compact dry powder agglomerates in turbulent flows within Eulerian–Lagrangian simulations. *Powder Technology* 348, 105–125. DOI: 10.1016/j.powtec.2019.03.009

Academic Distinctions

Wissenschaftspreis 2024 (Science Award), Co-recipient

Award for the best doctoral dissertation at Helmut-Schmidt University. Awarded by Freunde und Förderer der Helmut-Schmidt-Universität / Universität der Bundeswehr e.V., March 2024.

Internal Research Funding (IFF), 07.2023–06.2024

Competitively awarded one-year seed funding at Helmut-Schmidt University to prepare an external research grant application; selected through a written application and oral presentation.

DFG Research Grant, Co-Principal Investigator, 12.2025–11.2028

Deagglomeration in turbulent flows: Particle-resolved investigation for detailed insights and advanced modeling. Reference KH 653/3-1; project 573724408. Funding secured for a student research assistant position supervised at TUM, alongside contribution to supervision of the project's doctoral researcher in Hamburg.

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